sig01_fs48000.iq

Analysed 26 September 2026, 10:18 - 7.28 s of signal, 349,397 samples, sampling rate from user

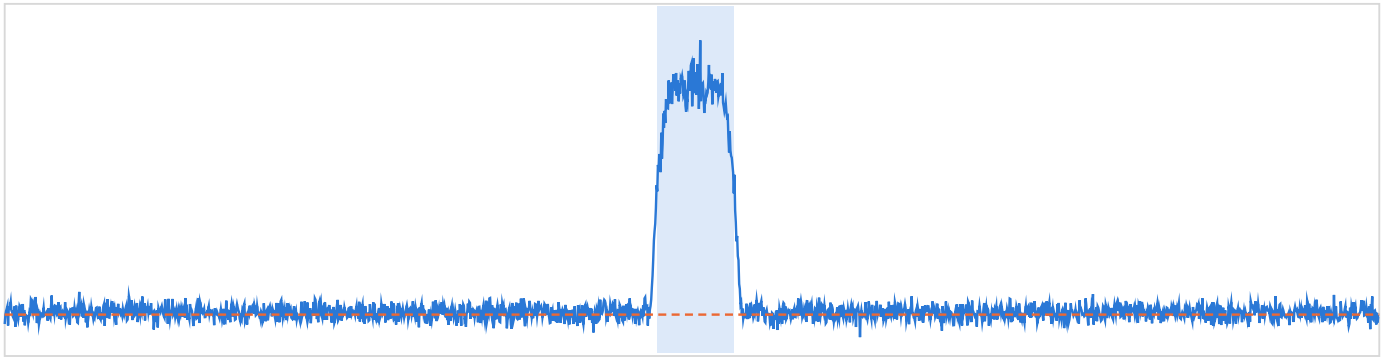
MODULATION	SYMBOL RATE	INTERLEAVER	ERROR CORRECTION	FRAMES	PAYLOAD
BPSK	**2,400 Bd**	**Block 28 x 37**	**Convolutional K=7 r=1/2**	**16**	**824 chars**

Signal parameters

file	**sig01_fs48000.iq**	classifier confidence	**0.94**
sampling rate (Hz)	**48000.0**	decision SNR (dB)	**12.3**
sampling rate source	**user**	frames found	**16**
duration (s)	**7.279**	frame length (bits)	**1084**
carrier offset (Hz)	**123.0**	phase/ambiguity	**rot 0 deg**
occupied bandwidth (Hz)	**2660.2**	interleaver	**Block 28 x 37**
spectral SNR (dB)	**11.6**	FEC	**Convolutional K=7 r=1/2 (171, 133)**
symbol rate (Bd)	**2400.05**	identification score	**1.0**
samples per symbol	**20.0**	frames with valid header	**16/16**
modulation	**BPSK**		

Every value above was measured from the samples themselves. The only thing supplied by the analyst is the sampling rate, and then only for a raw IQ file that carries no header.

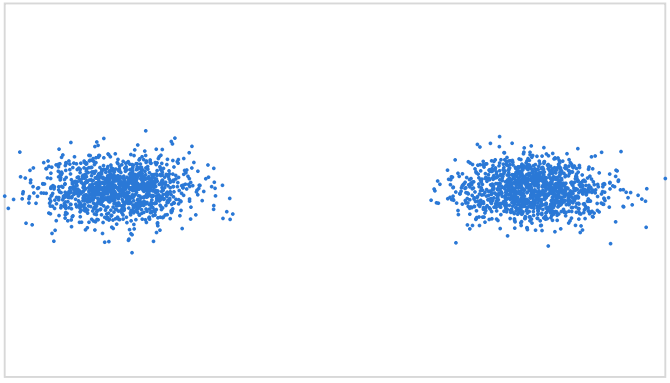
Power spectrum



frequency (Hz) power (dB)

Shaded: the occupied band, 2,660 Hz wide, centred +123 Hz from the middle of the recording. Dashed: the noise floor. Spectral SNR 11.6 dB.

Constellation after carrier recovery - BPSK



in-phase (I) quadrature (Q)

Decision summary

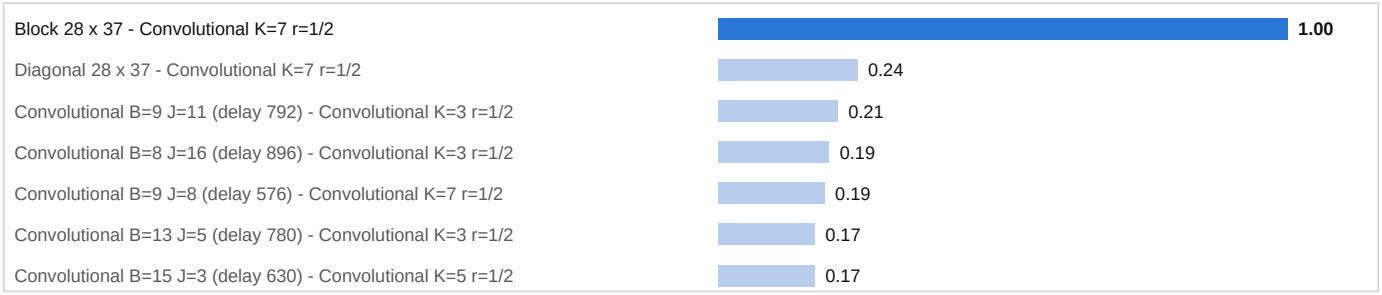
Modulation	**BPSK**
Symbol rate	**2,400.0 Bd**
Decision SNR	**12.3 dB**
Classifier confidence	**0.94**
Residual carrier offset	**-3.0 Hz**
Frames found	**16**
Frame length	**1084 bits**
Framing mode	**known preamble**

Left: the received symbols once the carrier has been recovered. Tight clusters on the ideal constellation points mean the signal is being read correctly.

Interleaver and error-correction identification

Interleaver	**Block 28 x 37**	Runner-up score	**0.244**
Error-correcting code	**Convolutional K=7 r=1/2 (171, 133)**	Hypotheses tested	**1,859**
Structure score	**1.000**	Coded block	**1036 bits (0 pad)**

Ranked hypotheses



structure score (1.0 = fits perfectly)

Every interleaver shape, convolutional setting and pseudo-random seed is applied, and each result is scored against every candidate code using algebraic structure: the parity relation of a convolutional code, Reed-Solomon syndromes and the Berlekamp-Massey error-locator length, unsatisfied LDPC parity checks. A wrong guess scores near zero, so the winner is decided by the margin rather than by a threshold we chose.

Interleaver	Error-correcting code	Coded bits	Pad	Score
Block 28 x 37	Convolutional K=7 r=1/2 (171, 133)	1036	0	1.000
Diagonal 28 x 37	Convolutional K=7 r=1/2 (171, 133)	1036	0	0.244
Convolutional B=9 J=11 (delay 792)	Convolutional K=3 r=1/2 (7, 5)	244	0	0.210
Convolutional B=8 J=16 (delay 896)	Convolutional K=3 r=1/2 (7, 5)	140	0	0.194
Convolutional B=9 J=8 (delay 576)	Convolutional K=7 r=1/2 (171, 133)	460	0	0.186
Convolutional B=13 J=5 (delay 780)	Convolutional K=3 r=1/2 (7, 5)	256	0	0.169
Convolutional B=15 J=3 (delay 630)	Convolutional K=5 r=1/2 (23, 35)	406	0	0.169
Convolutional B=12 J=7 (delay 924)	Convolutional K=5 r=1/2 (23, 35)	112	0	0.167

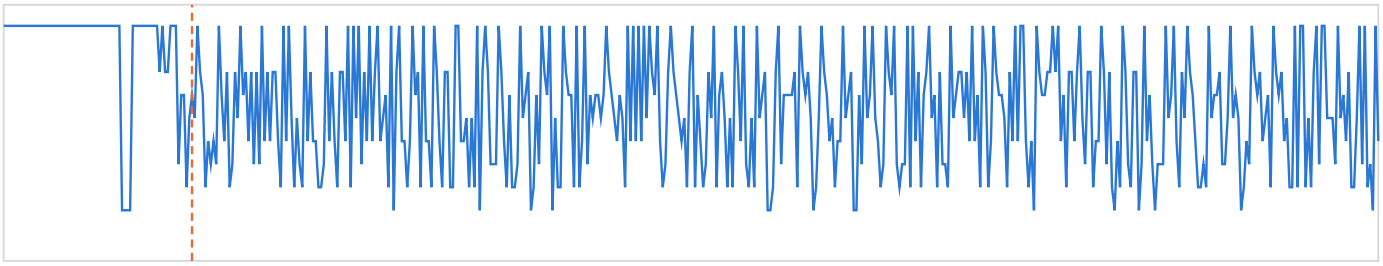
Decode quality

Frames decoded	**16/16**	RS symbols corrected	**-**
Channel BER before correction	**0.00e+00**	LDPC iterations (mean)	**-**

The channel introduced no bit errors the decoder had to repair at this signal-to-noise ratio: every frame re-encoded to exactly what was received.

Header and payload

Bit constancy across frames



bit position in the decoded frame fraction agreeing

Stacking the decoded frames and asking how often each bit position repeats separates the header (constant) from the payload (random). That puts the boundary near bit 70 without knowing the frame format. The sync word 1ACFFC1D was then located by cross-correlation in every frame.

Frame	Sync at bit	Match	Counter	Type	Length	Payload
0	0	1.00	0	0x5A	56	SIH 2026 // RF SIGNAL INTELLIGENCE TEST
1	0	1.00	1	0x5A	56	is payload was modulated, interleaved an
2	0	1.00	2	0x5A	56	the synthetic generator. If you can rea
3	0	1.00	3	0x5A	56	every stage of the receiver chain worke
4	0	1.00	4	0x5A	56	imation, modulation identification, demo
5	0	1.00	5	0x5A	56	erleaving, error correction and bitstrea
6	0	1.00	6	0x5A	20	- END OF MESSAGE --
7	0	1.00	7	0x5A	56	SIH 2026 // RF SIGNAL INTELLIGENCE TEST
8	0	1.00	8	0x5A	56	is payload was modulated, interleaved an
9	0	1.00	9	0x5A	56	the synthetic generator. If you can rea
10	0	1.00	10	0x5A	56	every stage of the receiver chain worke
11	0	1.00	11	0x5A	56	imation, modulation identification, demo

First 12 of 16 frames; the rest parsed identically.

Recovered message

SIH 2026 // RF SIGNAL INTELLIGENCE TEST TRANSMISSION. This payload was modulated, interleaved and FEC-encoded by the synthetic generator. If you can read this sentence, every stage of the receiver chain worked: parameter estimation, modulation identification, demodulation, de-interleaving, error correction and bitstream correlation. -- END OF MESSAGE -- SIH 2026 // RF SIGNAL INTELLIGENCE TEST TRANSMISSION. This payload was modulated, interleaved and FEC-encoded by the synthetic generator. If you can read this sentence, every stage of the receiver chain worked: parameter estimation, modulation identification, demodulation, de-interleaving, error correction and bitstream correlation. -- END OF MESSAGE -- SIH 2026 // RF SIGNAL INTELLIGENCE TEST TRANSMISSION. This payload was modulated, interleaved and FEC-encoded by

824 characters recovered from 16 frames. Every stage - parameter estimation, modulation identification, demodulation, de-interleaving, error correction and bitstream correlation - had to be right for this to be readable.

Analysis log

Each stage in the order it ran, with what it found. This is the audit trail behind every number in this report.

Stage	Detail
spectrum	occupied bandwidth 2660.2 Hz centred at +123.0 Hz, spectral SNR 11.6 dB
modulation	linear (PSK/QAM): no clean FSK tone clusters (FSK confidence 0.00); identified BPSK at 2400.05 Bd, decision SNR 12.3 dB
framing	16 frames of 1084 bits (known preamble); ambiguity resolved as 'rot 0 deg'
coding	interleaver: Block 28 x 37; FEC: Convolutional K=7 r=1/2 (171, 133) (score 1.000, 1859 hypotheses tested)
decode	estimated channel BER before FEC: 0.00e+00
correlation	sync word found in 16/16 frames; estimated header length 70 bits

Run detail - stage timings and framing

spectral analysis	**0.040 s**	coding identification	**0.178 s**	framing mod	**known preamble**	phase / ambiguity	**rot 0 deg**
symbol rate	**0.374 s**	decoding	**0.120 s**	frames	**16**	preamble	**C8B3A9F43893**
demodulation	**0.376 s**	correlation	**0.003 s**	frame length	**1084 bits**		
framing	**0.003 s**	total	**1.094 s**	block length	**1036 bits**		